Goal: Exploring Steepest Descent accelerations.

Reminders

1. **Heavy Ball Method** (f L-smooth, μ-convex)
   $x^{k+1} = x^k - \alpha\nabla f(x^k) + \beta(x^k - x^{k-1}), \quad \alpha, \beta > 0,\ x^0 = x^{-1} \in \mathbb{R}^n.$
2. **Nesterov optimal method** (f L-smooth, convex)
   $$\begin{cases} x^{k+1} = y^k - \frac{1}{L}\nabla f(y^k), \\ \tau_{k+1} = \frac{1}{2}\left(1 + \sqrt{4\tau_k^2 + 1}\right), & \tau_0 = 1 \text{ and } x_0 = y_0 \in \mathbb{R}^n. \\ y^{k+1} = x^{k+1} + \frac{\tau_k - 1}{\tau_{k+1}}(x^{k+1} - x^k). \end{cases}$$
3. **Nesterov optimal method for strongly convex functions** (f L-smooth, μ-convex)
   $$\begin{cases} x^{k+1} = y^k - \frac{1}{L}\nabla f(y^k), \\ y^{k+1} = x^{k+1} + \frac{\sqrt{L} - \sqrt{\mu}}{\sqrt{L} + \sqrt{\mu}}(x^{k+1} - x^k). & x_0 = y_0 \in \mathbb{R}^n. \end{cases}$$

Exercise 12.1

We consider the function

$$f : \mathbb{R}^2 \to \mathbb{R}, \quad f(x) := (x_1 - 3)^2 + k(x_0 - 2)^2$$

for $k = 10$ and $k = 0.01$.

1. Implement
   - Heavy ball method with $\alpha = \frac{4}{(\sqrt{L} + \sqrt{\mu})^2}$, $\beta = \frac{\sqrt{L} - \sqrt{\mu}}{\sqrt{L} + \sqrt{\mu}}$,
   - Nesterov's optimal method,
   - Nesterov's optimal method for strongly convex functions,

   where L denotes the Lipschitz constant of the gradient of f, and μ the modulus of convexity of f. Use $(0, 0)^\top$ as initial iterate. Terminate the loop as soon as $k \geq$ `maxiter` or $\|\nabla f(x^k)\|_2 \leq$ `epsilon`, with `maxiter` $= 1000$ and `epsilon` $= 10^{-5}$.
2. Draw a plot of the convergence behavior, plotting iteration number against $\log_{10}(f(x^k) - f(x^*))$. (Use the same figure, with different colors for each algorithm. Also add the gradient descent method using steplength $1/L$ and $2/(\mu + L)$.)
3. Discuss the results.

Exercise 12.2

Let $f : \mathbb{R}^n \to R$ be L-smooth and convex and let $x^0 \in \mathbb{R}^n$ be arbitrary. Assume that f admits a minimizer $x^* \in \mathbb{R}^n$. We consider the iteration

$$\begin{aligned} y^k &= x^k + \beta_k(z^k - x^k) \\ x^{k+1} &= y^k - \alpha_k \nabla f(y^k) \\ z^{k+1} &= z^k - \gamma_k \nabla f(y^k), \end{aligned}$$

with $\alpha_k = \frac{1}{L}$, $\beta_k = 1 - \frac{\omega_k}{\omega_{k+1}}$, $\gamma_k = \frac{\alpha_k}{\beta_k}$ for $\omega_{k+1} = \omega_k + \frac{k}{2}$ and $\omega_0 = 0$, $x^0 = y^0 = z^0$.

1. Show that $\beta_k = \frac{2}{k+1}$ for $k > 1$.

2. Show that the following equality holds:

$$\begin{aligned} &\|z^{k+1} - x^*\|_2^2 - \|z^k - x^*\|_2^2 \\ &= -\frac{\alpha_k^2}{\beta_k^2}\|\nabla f(y^k)\|_2^2 + \frac{2\alpha_k}{\beta_k^2}\langle \nabla f(y^k), x^* - x^{k+1}\rangle + \frac{2\alpha_k(\beta_k - 1)}{\beta_k^2}\langle \nabla f(y^k), x^* - x^k\rangle. \end{aligned} \tag{1}$$

3. Show that the sequence $\{\varphi_k\}_{k\in\mathbb{N}}$, $\varphi_k := (\frac{k}{2})^2(f(x^k) - f(x^*)) + \frac{L}{2}\|z^k - x^*\|_2^2$ is monotonically decreasing.
   Hint: Use the following inequality verified for $\eta_1 > 0$, $\eta_2 > 0$ and $\eta_2 - \eta_1 > 0$ in a substraction with (1)*.*

$$\begin{aligned} 0 \geq\ & \eta_2(f(x^{k+1}) - f(x^*)) - \eta_1(f(x^k) - f(x^*)) - \eta_2\frac{L\alpha_k^2}{2}\|\nabla f(y^k)\|_2^2 \\ &+ \eta_1\langle \nabla f(y^k), x^k - x^*\rangle + \eta_2\langle \nabla f(y^k), x^* - x^{k+1}\rangle, \end{aligned} \tag{2}$$

4. Show that the iterates $\{x_k\}_{k\in\mathbb{N}}$ satisfy

$$f(x^k) - f(x^*) \leq \frac{2L}{k^2}\|x^0 - x^*\|_2^2$$

   for all $k \in \mathbb{N}$.

5. Show that it yields the same sequences $(x^k)_{k\in\mathbb{N}}$ and $(y^k)_{k\in\mathbb{N}}$ as Nesterov's accelerated gradient method

$$\begin{aligned} y^k &= x^k + \tilde{\beta}_k(x^k - x^{k-1}) \\ x^{k+1} &= y^k - \tilde{\alpha}_k \nabla f(y^k), \end{aligned}$$

   with $\tilde{\alpha}_k = \frac{1}{L}$ and $\tilde{\beta}_k = \frac{k-1}{k+2}$ for all $k \in \mathbb{N}$.